

0x7DF J203404CONNECT

\$22 F1 98

PCM: FLASH_MODE



Tester Engineering Suite



Professional diagnostics,
properly **engineered.**

17,974

OEM FAULT CODES

20,592

PID FORMULAS

660

MODE-06 TIDs

15

OEM FAMILIES

40+

SEED-KEY ALGOS

Windows 7 – 11

.NET Framework 4.8.1

SAE J2534-1/2 PassThru

Developed in Australia

Professional diagnostics, properly engineered.

Tester Engineering Suite (TES) is the flagship Windows desktop platform from **Tester Present** — Australia's specialist automotive software-engineering consultancy. Designed for technicians, calibrators, and OEM engineering teams who require deeper access than commercial scan tools can provide, TES consolidates the scan tool, bench programmer, CAN analyser, binary editor, seed-key utility, and live datalogger into one integrated application.

Built on .NET Framework 4.8.1 and communicating through the SAE J2534-1/2 PassThru standard, TES works with every certified VCI on the market — Mongoose, OpenPort, CarDAQ, Tactrix, VXDIAG, OBDX Pro. Instead of juggling separate scan tools, bench programmers, CAN sniffers, bin editors, seed-key utilities, and reporting software, TES brings these capabilities together — backed by cryptographic integrity chains, validated flash profiles, and a code-signed update channel engineered for evidentiary, production, and regulated use.

// BUILT FOR

Engineers who need deeper access than off-the-shelf tools provide

UDS::0x27 SA

Workshop Technicians

Branded OEM workspaces, freeze-frame DTCs, emissions monitor tracking, PDF reports for customers, and ECU Birth Certificates for warranty work.

Calibrators & Tuners

Validated flash profiles for Ford Falcon, ZF 6HP26, Bosch ME7 BSL — with FEPS handling, pre/post-flash SHA-256 verification, and a binary diff tool.

Software Developers

Built-in multi-module J2534 emulator, ODX-to-routine creator, Seed-Key Sandbox, headless mode, and remote master/slave agent for off-bench work.

OEM Engineering Teams

30+ bench-grade engineering tools (Ctrl+Shift+A), CAN bit-timing analyzer, evidentiary HMAC-sealed certificates, and full UDS (ISO 14229) coverage.

ERING



Professional diagnostics, properly engineered.

A complete diagnostic platform

Scan tool, bench programmer, CAN analyser, binary editor, seed-key utility, and live datalogger — consolidated into a single, integrated application.

// 01 DIAGNOSTICS & CONNECTIVITY

01

General OBD-II Diagnostics

Modes 01–0A — live data, freeze frames, fuel-trim, O₂ sensor data, on-board test results, calibration IDs, pending and permanent DTCs.

SAE J1979 · ISO 15031-5

02

Full UDS (ISO 14229-1)

Every service that matters — sessions, security access, DataIdentifiers, routines, ECU-reset, transfer, communication control. Custom routine builder + NRC decoder.

\$10 \$14 \$19 \$22 \$27 \$2E \$31 \$34-37 \$3E

03

VIN Readout & Decoder

Mode 09 InfoType 02 and UDS \$22 F1 90 — WMI/VDS/VIS breakdown, ISO 3779 model-year decode, plant code, sequential number, check-digit validation, auto re-themes UI.

8 WMI-overlap conflicts corrected

04

Global & OBD-II Fault Codes

Full DTC management — Mode 03/07/0A for OBD-II; UDS \$19 sub-functions \$02/\$04/\$06 for OEM-specific codes. Freeze-frame, snapshot and extended data records.

Per-VIN integrity chain on every op

05

CAN & K-Line Bus Topographer

Active bus discovery sweeps the 11-bit and 29-bit CAN ID space and K-Line address range, identifies every responsive ECU, and renders the network topology.

HS-CAN · MS-CAN · SW-CAN · K-Line

06

Multi-Bus CAN Sniffer

HS-CAN (500k), MS-CAN (125k), SW-CAN (33.3k) auto-detect/manual; replay, filtering and a full CAN Bit-Timing Analyzer down to the segment-quantum level.

ISO 11898 / GMLAN / SAE J2284

// 02 LIVE DATA & ANALYSIS

07

1000+ OEM PID Definitions

Pre-decoded Mode 22 databases for Ford, GM, Honda, Nissan, Mazda, Subaru, PSA/Citroën, Daimler — plus a custom PID builder with full formula editor.

20,592 scaling formulas underpin

08

Live Datalogger & Signal Analyser

Simultaneous multi-PID capture at the highest rate the ECU supports — real-time plotting, cursor-pair measurement, frequency-domain analysis, CSV export.

Bench-grade · cursor & statistical

09

Relative Compression & Misfire

RPM-ripple compression analysis via PID 0x0C while cranking — finds weak cylinders without pulling plugs. Plus per-cylinder Mode 06 misfire counters.

660 Mode-06 TIDs pre-labelled

Programming, flash & specialty tools

// 03 PROGRAMMING, FLASH & SERVICE

SW-CAN 33.31

10

ECU Flash Reprogramming

Read, erase, write firmware for the Ford Falcon family (BA/BF/FG/FGII/FG-X), ZF 6HP26 TCM, and Bosch ME7 BSL bootmode. FEPS-aware, SHA-256 verified.

Validated profiles – bench-tested

11

40+ Seed-Key Algorithms

Ford CAN-Generic v2003, Ford PATS, and UDS \$27 coverage across Ford, JLR, VW, GM, Honda, Nissan, Mazda, Subaru, Daimler, PSA/Stellantis, Renault — plus Sandbox.

400+ Ford named-key dictionary

12

Ford Routine Runner

One-click execution of OEM service routines — throttle/SAS/EPB resets, learn procedures, calibration commits, oil-life resets, immobiliser sync.

Audit-logged on every execution

13

Ford As-Built Data Editor

Read and edit factory As-Built configuration blocks — option-coding, retrofits, region changes, immobiliser configuration, module-swap data carry-over.

Per-module diff · snapshot restore

14

PATS Programming — BA & FG

Passive Anti-Theft System programming for BA/BF and FG/FGII/FG-X Falcon — key learning, PCM↔ICC sync, transponder enrolment, emergency unlock workflows.

Validated on hardware

15

ODX → Routine Creator

Import any ODX-D diagnostic description file and the suite generates an executable J2534 routine on the fly — fastest path from OEM spec to bench-tested procedure.

ISO 22901-1 · ODX-LINK output

// 04 SPECIALTY & BENCH TOOLS

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16

30+ Advanced Diagnostic Tools

Ctrl+Shift+A opens the engineering library — relative compression, ground-loop detector, CAN bit-timing analyzer, ELM327 comparator, DTC timeline, seed-key sandbox.

Extended every release

17

Built-in J2534 Emulator

Complete multi-module ECU bus simulator — Ford FG/EEC-VI/TCM responds to UDS, OBD-II, and security access with real seed-key algorithms. Headless mode for CI.

Develop without hardware

18

Remote Master/Slave Agent

Drive a J2534 device on the other side of the world — encrypted WebSocket relay, full feature parity with local mode. Flash, diagnose, sniff over the internet.

Audit-logged · signed envelopes

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15 OEM families, indexed

Every fault code, PID formula, security algorithm and on-board test ID is bucketed by OEM with an automatic SAE J2012 Generic fallback. Coverage figures reflect TES v1.0.5-2026 and are continuously expanded through the secure update channel.

OEM FAMILY	BRANDS COVERED	DTCs	FORMULAS	DIDs
Ford	Ford, Lincoln, Mercury	3,138	3,420	2,180
GM	Chevrolet, Buick, Cadillac, Pontiac, Saturn, Oldsmobile, GMC, H...	1,802	2,310	1,605
VAG	VW, Audi, Skoda, SEAT, Porsche	1,431	3,180	1,872
BMW	BMW, MINI	1,205	1,910	1,140
Mercedes	Mercedes-Benz, Smart	1,142	1,580	990
Chrysler / Stellantis	Chrysler, Dodge, Jeep, Ram	980	1,260	820
Toyota	Toyota, Lexus, Scion	895	1,485	760
Honda	Honda, Acura	812	1,140	630
Nissan	Nissan, Infiniti	762	1,025	580
Hyundai / Kia	Hyundai, Kia	724	935	510
Mazda	Mazda	485	620	340
Subaru	Subaru	412	540	290
Jaguar / Land Rover	Jaguar, Land Rover	385	490	240
Volvo	Volvo	298	385	180
Generic (SAE J2012)	Any OBD-II compliant vehicle	3,503	312	—
TOTAL	15 OEM families · ~60 brands	17,974	20,592	12,137

Continuously expanded release over release · Delivered to licensed installations through the code-signed update channel.

Per-OEM branded workspaces

Every OEM family ships with its own gradient banner, accent colour, and brand glyph. VIN-decode re-themes the workspace automatically so the technician's UI matches the vehicle in front of them. Includes a dedicated VAG diagnostic workspace with Security Access, Long Coding, Adaptation, Immobilizer wizard, and 14 standard VAG diagnostic addresses pre-loaded.

Engineered like infrastructure, not a scan tool

Validated flash profiles, cryptographic integrity chains, and a code-signed update channel — every layer of the platform engineered for evidentiary, production, and regulated use.

VALIDATED FLASH PROFILES

Every supported ECU has a hand-validated flash profile — not “should-work” speculation from a generic UDS template. Each profile encodes the exact bootmode handshake, memory-map constraints, security-access requirements, and recovery behaviour.

- Pre-flash SHA-256 verification against the expected calibration
- FEPS / bootmode handshake validation before any erase or write
- Battery-voltage and supply-current supervision throughout
- Transaction-level abort, rollback and re-entry on transient fault
- Post-flash CVN and Cal-ID readout, written into the per-VIN chain

CODE-SIGNED UPDATE CHANNEL

TES software and OEM coverage data are delivered through Tester Present's signed update channel. Every release is code-signed; every coverage-library update carries an integrity stamp verified before installation. The platform fails closed on tampering.

- Authenticode-signed Windows executables and installers
- Integrity-stamped OEM coverage releases, per-licence
- Per-installation update authorisation via LicenseServer
- Offline-grace caching for intermittent connectivity sites
- Rollback retained for one full minor version

SHA-256 TAMPER-EVIDENT VIN HISTORY

Every action TES performs — flash, DTC clear, security access, As-Built edit, PATS sync, routine execution — is appended to a per-VIN SHA-256 hash chain. A single edit anywhere makes the chain stop validating; the verify-chain tool pinpoints the line.

- Local-only storage — never uploaded
- Auto-stamped on every PDF export and certificate
- Separable per workshop / per technician
- Forward-migrated through every schema change
- Cryptographic linked-list integrity

HMAC-SEALED ECU BIRTH CERTIFICATE

Standalone evidentiary record documenting firmware hash, calibration ID, software/hardware part numbers, VIN, technician, timestamp and licence at a specific moment. Sealed with HMAC-SHA256 — any post-hoc edit invalidates the certificate.

- Warranty work — proves the OEM shipped that calibration
- Recall sign-off — cryptographic proof the update landed
- Fraud investigations — tamper-evident chain-of-custody
- Legal disputes — engineering evidence, not screenshots

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Speaks every dialect that matters

0x430 STEERING

A complete, fully-implemented protocol stack — not a thin wrapper around third-party libraries.

MS-CAN @ 125k

// DIAGNOSTIC STANDARDS

- > UDS · ISO 14229-1
- > SAE J2534-1/2 PassThru
- > OBD-II · SAE J1979
- > ISO 15031-5
- > J1979-DA · Mode-06
- > SAE J2012 · DTCs
- > ISO 3779 · VIN decode
- > ISO 22900-2 · D-PDU
- > ODX-D · ISO 22901-1

// TRANSPORT & PHYSICAL

- > ISO 15765 CAN (11/29-bit)
- > ISO-TP single + multi-frame
- > HS-CAN · 500 kbps
- > MS-CAN · 125 kbps
- > SW-CAN · 33.3 kbps GMLAN
- > ISO 9141-2
- > ISO 14230 KWP2000
- > VAG KWP1281
- > SAE J1850 VPW / PWM

// OEM BOOTMODE / FLASH

- > Bosch ME7 BSL
- > Ford FEPS (18 V FE)
- > ZF 6HP26 TCM
- > Simos18 LZSS
- > Ford VBF / LZSS
- > Ford As-Built data
- > PATS BA · PCM-ICC sync
- > PATS FG · PCM-ICC sync

// VAG-SPECIFIC

- > UDS 0x27 · 16 PINs
- > VGS & VGS-Ext algorithms
- > VW-SA2 sec access v2
- > Long Coding bytes
- > Adaptation channels
- > Component Protection
- > SCN coding
- > 14 standard VAG addrs

// HARDWARE INTERFACES

- > OBDX Pro
- > Ford VCM2
- > Drew Tech Mongoose
- > Tactrix OpenPort 2.0
- > Drew Tech CarDAQ-Plus
- > VXDIAG VCX
- > Bosch Mastertech
- > ELM327 (basic only)

// REPORTING & EVIDENCE

- > Multi-page PDF export
- > MANIFEST.json + hashes
- > Certificates of Authenticity
- > ECU Birth Certs (HMAC)
- > Per-VIN SHA-256 chain
- > Verify-chain forensic tool
- > Audit-logged routines
- > Insurance / legal use

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0x430 STEERING

Two options — no automatic subscriptions

Choose the term that fits your operation. Both tiers unlock the complete suite — every feature, every supported OEM, every flash profile, every advanced tool. Hardware-fingerprint-bound. No per-feature unlocks. No background billing.

12-MONTH LICENCE

Annual access · renewed at user discretion

AUD \$800

 / 12 months

- ✓ Full unrestricted access to all functionality
- ✓ 12 months from the date of purchase
- ✓ Technical & diagnostic support
- ✓ Updates within the licensed version
- ✓ Hardware-fingerprint-bound activation

Not a subscription. No auto-renewal. You decide when (or if) to renew.

LIFETIME LICENCE ★ BEST VALUE

One-off fee · lifetime updates · priority support

AUD \$2,500

 one-off

- ✓ Full unrestricted access for life
- ✓ Lifetime updates — every future version
- ✓ Priority technical & diagnostic support
- ✓ Early access to new OEM coverage
- ✓ Hardware-fingerprint-bound activation

Pay once. Use forever. Every future major version included.

// BOTH TIERS INCLUDE

17,974 OEM DTCs · 20,592 PID scaling formulas · 660 Mode-06 TIDs · 15 OEM families · 40+ seed-key algorithms · 30+ advanced diagnostic tools · ECU flash for Ford Falcon BA/BF/FG/FGII/FG-X, ZF 6HP26 TCM, Bosch ME7 BSL · PATS BA/FG · As-Built editor · ODX → routine creator · live datalogger · tamper-evident SHA-256 VIN history · HMAC-sealed ECU Birth Certificates · remote master/slave agent · PDF reporting throughout.

// REQUEST A LICENCE OR TECHNICAL EVALUATION

All enquiries answered by an engineer within one business day. For volume sales, OEM engineering teams, and site-wide deployments, get in touch:

EMAIL

contact@testerpresent.com.au

WEB

tester.engineering

PARENT

testerpresent.com.au


https://tester.engineering